

Chapter 1 (Advanced Python Concepts and Best Practices)

1. Explain different types of arguments in python function with example.
2. Explain lambda expressions. How do you use lambda expressions in filter , map and reduce. Explain with example.
3. What are decorators and generators in python. Explain with example.
4. Explain exception handling in python with try,except and finally. Also show customized exceptions in python.
5. Explain python logging with different types of logging levels. Also write a program to write exception information to the log file.
6. Why do you need Assertion in python. Differentiate between exception handling and assertions.
7. Why do you need `__super__` function in python ? Explain with example.
8. Explain different types of variables and methods in python with examples.
9. Differentiate between IS-A relationship and HAS-A relationship in python with example.
10. Explain different types of inheritance in python with examples.
11. Create a class named Student with attributes such as name, age, and grade. Write a program to create multiple objects of the class and display their details using a method.Your program should contain constructor,instance variable and instance methods.
12. Define a class Dog with attributes name and breed. Include a method bark() that prints a sound. Create objects of the class and demonstrate how the method works.Your program should contain constructor,instance variable and instance methods.
13. Create a class Rectangle that has attributes length and width. Implement methods to calculate the area and perimeter of the rectangle and display the results.Your program should contain constructor,instance variable and instance methods.
14. Create a class with different number of attributes and behavior. Your class must contain constructor. Create two objects and initialize with different values. Call all the methods you create. Define instance method and instance variable with example.
15. Explain OOP in Python. Create a class DataProcessor with methods for loading, cleaning, and saving data.
16. Write short notes on:
 - OOP
 - Iterators Vs Generators
 - Modules and packages
 - *args and **kwargs

Chapter 2 (Data Sources and APIs)

1. Explain different types of datas.
2. How do you read and write pdf and excel file ? Explain with python example.

3. Explain types of databases(relational and non relational) in detail with their characteristics. How do you access these databases ? Also compare with python ORM.
4. Define API. Also explain its communication flow.
5. Differentiate between REST Vs SOAP API.
6. Describe REST APIs and their HTTP methods.
7. Explain the steps to scrap data from public web page using BeautifulSoup.
8. The website 'suryapatro.com' contains 100 astrologer records with the attributes id, name,address, and affiliation. The attributes id, name, and address contain single values, while affiliation contains multiple values. Extract 30 records ranging from id 50 to 80 using the Requests and BeautifulSoup libraries.
9. Write Short note on:
 - A. Chunking
 - B. Lazy Evaluation

Chapter 3(Advanced Data Wrangling and Transformation)

1. Explain Merging, Joining , Reshaping and Pivoting in pandas with example.
2. How do you handle missing values in in python ?
3. How do you handle categorical, timeseries data in python ? Explain with example.
4. Explain feature transformation, encoding, and scaling operations.
5. Explain the various components involved in a data cleaning pipeline.
6. Design a reusable data-cleaning pipeline class (Assume necessary data if needed).
7. Write short note on:
 - Memory Optimization.

Chapter 4 (Applied Statistics and Exploratory Analysis)

1. Explain covariance, skweness, correlation and kurtosis with example. (Use necessary python package)
2. What is exploratory data analysis (EDA) ? How do you automate EDA work flow ? Explain in detail with codes. Your workflow should contain all necessary function.
3. What is Auto-EDA? Compare ydata-profiling and sweetviz libraries.
4. Write short note on:
 - a. Linear Regression
 - b. Trend Analysis
 - c. Discriptive Vs Inferential Statistics
 - d. Hypothesis testing

Chapter 5 (Data Visualization and Storytelling)

1. Explain principles of effective visualization.
2. Compare Matplotlib, Seaborn, and Plotly. Create box plot and pair plot using Seaborn.
3. Write code for an interactive Plotly dashboard with trends, pie chart, and bar chart.
4. Compare static vs interactive visualizations. When to use Plotly over Matplotlib?
5. Explain dashboard thinking. Design a notebook dashboard with KPIs, trends, and comparisons.
6. Explain the importance of colors in visualization. A city authority collects hourly traffic volume, accident count, accident severity, location (junction-wise). Which visualization approach best reveals high-risk zones and peak hours? Explain with pros and cons of visualization approach you suggest.
7. Write Short notes on:
 - a. Create subplots with line, bar, and histogram
 - b. Key principles of data storytelling
 - c. Visualization driven insight generation.

Chapter 6 (Data Engineering and Automation)

1. Design and implement ETL pipeline of your own .
2. Describe logging, monitoring and error handling in ETL pipelines.
3. Explain about different data storages and retrieval strategies in data pipelines.
4. What are DAGs in data pipelines? Explain task dependencies.
5. Describe automated report generation. Write a function that generates HTML report.
6. A growing online education platform wants to track student activity, test performance, and revenue. Generate daily and monthly analytics reports automatically. Deliver reports in Excel (operations), HTML (dashboard), and PDF (management) formats. Eliminate manual reporting work
7. Write short notes on:
 - a. Data Science Vs Data Engineering.
 - b. ETL Vs ELT
 - c. Scheduler Vs CRON